Mechanistic Hypothesis Search — Final Report

Target Disease: MYH7-Related Skeletal Myopathy (Mendelian)

Hypothesis: `rod_srx_energetics` — "Rod variants cause weakness by releasing myosin heads from the super-relaxed state, raising ATP use and causing fatigue"

KB Status: EMERGING | Search date: 2026-10-16 | Iterations: 5

Executive Judgment

Verdict: PARTIALLY SUPPORTED.

The *core energetic claim* of this hypothesis is directly and independently supported: MYH7 rod/light-meromyosin (LMM) variants reduce the sequestered myosin **super-relaxed (SRX)** state and raise **basal myosin ATP consumption**. This is demonstrated in the **R1500P Laing distal myopathy mouse** model (PMID: 38690726) and, critically, in **patient biopsies** carrying MYH7/MYH2 LMM mutations (PMID: 37788100). The mechanism is biologically plausible given MYH7's genetic constraint signature — the gene is extremely missense-intolerant (gnomAD mis_z = 7.38) but loss-of-function–tolerant (pLI ≈ 0, LOEUF = 0.66), consistent with an **altered-function** (not haploinsufficiency) disease model that an SRX-disrupting dominant missense mutation would produce.

However, three substantial caveats prevent a stronger verdict. **First, the hypercontractility component is model-organism-specific.** The R1500P mouse is explicitly hypercontractile, but human patient myofibers show raised ATP demand *without* altered contractility — the authors of the human study frame the mechanism as energetic rather than contractile (PMID: 37788100). **Second, the decisive causal step — raised ATP demand → fatigue/weakness — is**

not directly measured in patients. No MYH7-specific in-vivo energetics-to-weakness study (e.g., ^{31}P-MRS in genotyped patients) was located as of the search date. **Third, SRX loss is a convergent, non-MYH7-specific node.** The same SRX-loss-plus-raised-resting-ATP signature appears in nebulin (NEB)-related nemaline myopathy — a *thin-filament* disease (PMID: 36528760) — so SRX disruption is a shared downstream consequence of diverse sarcomeric lesions rather than a MYH7-specific primary driver.

Finally, a well-established **competing mechanism** exists for the very same variant: Buvoli et al. showed R1500P and L1706P act as **dominant-negative** mutations that form aggregates sequestering wild-type myosin and perturb thick-filament assembly (PMID: 22155079). Because this aggregation/assembly-defect mechanism co-occurs for R1500P and could lie upstream of or parallel to SRX loss, the **causal primacy** of the energetic arm within human disease remains unresolved. The hypothesis is therefore best described as a mechanistically sound, well-evidenced *component* of MYH7 rod-myopathy pathophysiology whose contribution to human weakness, and whose primacy relative to structural/proteostatic defects, has not yet been established.

Key Findings

F001 — Rod (LMM) MYH7 variants reduce the super-relaxed state and raise basal ATP use (SUPPORT)

This is the strongest direct evidence for the hypothesis, resting on two independent primary studies using complementary human and mouse systems.

In the **R1500P Laing distal myopathy mouse** (PMID: 38690726), isolated muscle, myofibril, and myofiber contractility assays revealed "*a conformational shift of the myosin heads toward the more reactive disordered relaxed (DRX) state, causing hypercontractility and greater ATP consumption.*" The model also showed an increased actin–myosin detachment rate, fatigue, and weakness. Pharmacologically, **MYK-581** — a DRX-reducing (SRX-stabilizing) small-molecule myosin modulator — improved running capacity in the mice and corrected the elevated DRX fraction in *patient* myofibrils, providing a rescue/perturbation arm that strengthens the causal interpretation of the energetic defect.

In **patient biopsies** carrying MYH7/MYH2 LMM mutations (PMID: 37788100), combined mantATP-chase biochemistry and X-ray diffraction showed that "*basal myosin ATP consumption was increased and the myosin super-relaxed state was decreased in vivo.*" In-silico modeling in the same study indicated disruption of coiled-coil structure and filament packing. The convergence of a genetically faithful mouse and human patient tissue on the same energetic signature — SRX down, basal ATP up — is the firmest pillar of the hypothesis.

F002 — Human LMM-mutant myofibers show raised ATP demand WITHOUT altered contractility (QUALIFY)

The human study directly qualifies the "hypercontractility" language carried over from the mouse. In the same patient biopsies (PMID: 37788100), myofiber mechanics experiments found that "*myofiber contractility was not affected*," and the authors explicitly framed the lesion as "*increasing myosin head ATP demand in the filaments, rather than affecting contractility.*" This is a meaningful divergence from the R1500P mouse, where heads shift to DRX and muscle is explicitly hypercontractile. The **energetic / SRX-loss arm is concordant across species**, but the **hypercontractility arm is mouse-specific** and not yet demonstrated in human tissue. The seed hypothesis's phrase "in the R1500P mouse, muscle is hypercontractile" is accurate, but should not be generalized to humans.

F003 — Competing/complementary mechanism: coiled-coil, thick-filament assembly, and aggregation defects (COMPETING)

A classical mechanistic framework for MYH7 LMM disease holds that these mutations act through structural and proteostatic routes rather than (or in addition to) energetics. The *Myosinopathies* review (PMID: 22918376) states that LMM mutations produce "*altered biochemical and biophysical properties of the light meromyosin, which is essential for thick filament assembly*," and that "*In myosin storage myopathy protein aggregates are formed by accumulation of myosin beneath the sarcolemma and between myofibrils.*" The human SRX study (PMID: 37788100) itself reported that coiled-coil structure and filament packing were commonly disrupted in vitro. These assembly/aggregation defects could cause weakness independently of, or upstream of, SRX loss, making them a genuine competing mechanism rather than merely a downstream readout.

F004 — SRX destabilization + raised resting ATP is a convergent mechanism across structurally distinct genes (QUALIFY — reduces MYH7-specificity)

The SRX-loss/energetic-stress phenotype is not unique to MYH7. Nebulin (**NEB**)-related nemaline myopathy — a **thin-filament** gene, not myosin — also impairs the super-relaxed state and raises resting ATP use: "*the myosin stabilizing conformational state, known as super-relaxed state, was significantly impaired, inducing an increase in the energy (ATP) consumption of resting muscle fibres from NEB-NM patients*" (PMID: 36528760). The same assay pipeline (mantATP chase, X-ray diffraction, LC-MS/MS proteomics) shows SRX dysregulation in type 2 diabetes muscle (PMID: 40295335, where SRX is *increased* and ATP demand *decreased* — the opposite direction). Foundational work established that "*the structure of the thick filament modulates the nucleotide turnover rates of myosin in relaxed fibers*" and links this to thermogenesis (PMID: 19966283). SRX is therefore a **convergent hub** perturbable by many upstream lesions, which weakens any claim that SRX loss is a MYH7-specific primary driver.

F005 — Methodological caveat: structural and biochemical SRX readouts are not strictly equivalent (QUALIFY)

The two seed papers rely on X-ray diffraction order (structural OFF/ON) *and* mantATP chase (biochemical SRX/DRX) and treat them as reporting the same underlying SRX loss. Two methodological studies caution against this. In relaxed porcine myocardium, "*the structural OFF to ON transitions are not equivalent to the biochemically defined SRX to DRX transitions*" (PMID: 36696446). Separately, a re-examination of the mantATP displacement assay found that "*the commonly employed mantATP displacement assay was unable to quantify the population of the SRX state*," describing an SRX/IHM/ordered-filament "paradox" (PMID: 38103642). These do not refute the seed findings, but they mean the two readouts should be interpreted as correlated proxies rather than a single ground truth.

F006 — gnomAD constraint supports altered-function, not haploinsufficiency (SUPPORT, computational)

A gnomAD GraphQL constraint query (GRCh38, retrieved 2026-10-16; artifact `data/gnomad_MYH7_constraint.json`) returned: MYH7 **pLI = 6.6 × 10^{-21} (≈ 0)**, **LOEUF = 0.66**, observed/expected LoF = 129/225 = 0.57, lof_z = 5.45, but **missense mis_z = 7.38** (extreme; the conventional significance threshold is ≈ 3.09). Synonymous z = 0.69 (null), confirming calibration. The signature — strong missense intolerance with relative LoF tolerance — is consistent with a disease driven by **altered-function** / **dominant missense** variants (such as

an SRX-disrupting rod substitution) rather than loss of one allele. This supports the *premise* of the hypothesis (that rod missense variants exert a gain/change-of-function effect) without adjudicating between SRX-energetics and dominant-negative aggregation.

F007 — Competing mechanism: rod proline variants act as dominant-negative, sequestering wild-type myosin (COMPETING)

Buvoli et al. (PMID: 22155079) showed that R1500P and L1706P MPD1 mutations inhibit myosin self-assembly in non-muscle cells but do not prevent sarcomere incorporation; L1706P mislocalizes to the bare zone. Critically, "*Both mutations also form aggregates consisting of cytoplasmic vacuoles surrounding paracrystalline arrays and amorphous rod-like inclusions that sequester WT myosin*," and aggregates also formed in transgenic R1500P nematodes. Because **R1500P is the very variant** later shown to shift heads to DRX in the mouse (PMID: 38690726), the aggregation/dominant-negative mechanism and the SRX-energetic mechanism are entangled for the same allele. The clinical series of Lamont et al. (PMID: 24664454, 88 patients/21 families; distal weakness and footdrop in 81%, frequent de novo mutations) states plainly that "*The mechanisms by which different phenotypes are produced by mutations in MYH7, even in the same region of the gene, are not known.*"

F008 — ClinVar landscape is cardiomyopathy-dominated; skeletal rod-myopathy under-represented (data-gap context, computational)

An NCBI ClinVar esearch (db=clinvar, 2026-10-16; artifact `data/clinvar_MYH7_counts.json`) returned MYH7[gene] = 6011 records, 1479 pathogenic/likely-pathogenic. Cardiomyopathy-associated records = 5070 (~84%), versus skeletal-myopathy free-text matches: Laing = 396, distal myopathy = 414, myosin storage = 691 (controlled "myosin storage myopathy"[dis] = 397; "Laing distal myopathy"[dis] mapped to 0 — a vocabulary artifact confirmed by the 396 free-text hits). The evidence base for the *skeletal* phenotype that this hypothesis addresses is comparatively sparse, which constrains cohort-scale validation.

Mechanistic Model / Interpretation

The hypothesis proposes the following causal chain. Strength of each link is annotated.

```
MYH7 rod (LMM) missense variant  [STRONG: genetics + constraint, F006]
       │
       ▼
Perturbed coiled-coil / thick-filament backbone packing   [STRONG in vitro, F001/F003]
       │
       ├──────────────► Aggregation / assembly defect, WT sequestration  [COMPETING, F007]
       │                (dominant-negative; parallel or upstream route)
       ▼
Fewer heads in sequestered SRX / shift toward DRX   [STRONG: mouse + human, F001]
       │
       ▼
Increased basal (resting) myosin ATP consumption   [STRONG: mouse + human, F001]
       │
       ├── Mouse: hypercontractility  [SPECIES-SPECIFIC, F002]
       ├── Human: NO contractility change  [QUALIFIES, F002]
       ▼
Energetic stress / raised ATP demand at rest   [MODERATE: measured ex vivo, not in-vivo human]
       │
       ▼  ◄── INFERRED, NOT DIRECTLY MEASURED IN PATIENTS
Fatigue and muscle weakness   [WEAK causal link in humans; clinical phenotype real, F007]
```

The chain is strong from variant to "raised basal ATP consumption": two independent systems agree, and a myosin modulator (MYK-581) rescues both the DRX elevation and running capacity in the mouse, providing perturbational support. The chain **weakens decisively at the final step** — the leap from measured ex-vivo energetic stress to in-vivo clinical fatigue/weakness is inferred, not demonstrated, in MYH7 patients. The chain is also **branched**: the same backbone perturbation that lowers SRX can independently produce aggregates that sequester wild-type myosin (F007), and SRX loss itself is reachable from unrelated genes (F004). The hypothesis thus describes a *real and well-measured component* of pathophysiology whose standalone sufficiency and causal primacy in humans are unestablished.

Causal link	Evidence tier	Status
Rod variant → altered (dominant) function	Genetic constraint (F006)	Established
Rod variant → backbone/coiled-coil perturbation	In vitro + in silico (F001, F003)	Established
Backbone perturbation → SRX loss / DRX shift	Mouse + human biopsy (F001)	Established
SRX loss → raised basal ATP consumption	Mouse + human biopsy (F001)	Established
SRX loss → hypercontractility	Mouse only (F002)	Species-specific / Emerging
Raised ATP demand → fatigue/ weakness in patients	Not directly measured (F002)	Speculative / Inferred
SRX loss is MYH7-specific primary driver	Convergent across NEB, T2DM (F004)	Contradicted as *specific*; it is a shared hub
Aggregation/dominant-negative as competing route	Mouse/nematode/cell (F007)	Established competing mechanism

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic claim tested	Key finding	Disease subtype/ context
PMID: 38690726	Model organism (mouse)	**Supports**	Rod variant → DRX shift, ↑ATP, hypercontractility, fatigue	Heads shift to DRX; ↑ATP consumption; MYK-581 rescues running + corrects DRX in patient myofibrils	R1500P Laing distal myopathy
PMID: 37788100	Human clinical (biopsy)	**Supports / Qualifies**	SRX loss + ↑basal ATP in vivo; contractility	SRX ↓, basal ATP ↑ in vivo; contractility NOT affected; coiled-coil/ packing disrupted	MYH7/MYH2 LMM patients
PMID: 22918376	Review	**Competing**	LMM mutations → assembly defect + aggregation	LMM properties essential for thick-filament assembly; aggregates in myosin storage myopathy	MYH7 myosinopathi
PMID: 36528760	Human + model	**Qualifies**	Is SRX loss MYH7-specific?	NEB (thin-filament) mutations also impair SRX and ↑resting ATP	NEB nemaline myopathy
PMID: 19966283	In vitro / physiology	**Supports (foundational)**	Thick-filament structure ↔ SRX turnover	Filament structure modulates relaxed myosin nucleotide turnover;	Skeletal muscl (general)

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic claim tested	Key finding	Disease subtype/ context
				thermogenesis link	
PMID: 40295335	Human clinical	**Qualifies (context)**	SRX/energetics dysregulation generality	T2DM muscle: SRX *increased*, ATP demand *decreased* (opposite direction)	Type 2 diabete
PMID: 36696446	In vitro (porcine)	**Qualifies (method)**	Are structural OFF/ON ≡ biochemical SRX/ DRX?	Structural OFF→ON ≠/ biochemical SRX→DRX	Cardiac myocardium
PMID: 38103642	In vitro	**Qualifies (method)**	Does mantATP assay quantify SRX?	mantATP displacement unable to cleanly quantify SRX population	Cardiac HMM
PMID: 22155079	In vitro + model	**Competing**	Do rod prolines act dominant-negative?	R1500P/L1706P form aggregates that sequester WT myosin; assembly inhibited	MPD1 (R1500P L1706P)
PMID: 24664454	Human clinical (cohort)	**Context**	Genotype–phenotype mechanism	88 pts/21 families; distal weakness 81%; mechanisms "not known"	MYH7 distal myopathy
gnomAD v4 (computational)	Computational	**Supports (premise)**	Altered-function vs haploinsufficiency	mis_z=7.38; pLI≈0;	MYH7 gene-level

Citation	Evidence type	Supports/ Refutes/ Qualifies/ Competing	Mechanistic claim tested	Key finding	Disease subtype/ context
				LOEUF=0.66 → altered-function	
NCBI ClinVar (computational)	Computational	**Context (data gap)**	Skeletal vs cardiac evidence base	~84% cardiomyopathy; skeletal sparse	MYH7 variant landscape

Data and Tool Use

Source	Accession / URI	Access status	Query / filters	Resolved & relevant?
PubMed	Multiple PMIDs (17 papers reviewed)	**Accessed**	MYH7 rod/LMM, super-relaxed state, Laing distal myopathy, energetics, DRX, aggregation	Yes — directly on-topic
gnomAD (v4, GRCh38)	GraphQL constraint API; gene MYH7	**Accessed**; artifact `data/gnomad_MYH7_constraint.json` (retrieved 2026-10-16)	Gene-level constraint: pLI, LOEUF, mis_z, lof_z, syn_z	Yes — resolved; relevant to altered-function premise
NCBI ClinVar	eutils esearch, db=clinvar	**Accessed**; artifact `data/clinvar_MYH7_counts.json` (2026-10-16)	`MYH7[gene]`; disease sub-counts (cardiomyopathy, Laing, distal, myosin storage)	Yes — resolved; relevant to evidence-base skew; note vocabulary artifact ("Laing distal myopathy"[dis]→0)
GenCC	gencc.org	**Could not be verified** / **inaccessible** during run	Attempted gene-disease validity lookup for MYH7 skeletal myopathy	Not resolved — labeled unverified
ClinGen	clinicalgenome.org	**Could not be verified** / **inaccessible** during run	Attempted dosage/validity curation lookup	Not resolved — labeled unverified

Analysis inventory (input → method → output):

1. **gnomAD constraint retrieval** — Input: MYH7 gene symbol → Method: GraphQL constraint query, GRCh38 → Output: `data/gnomad_MYH7_constraint.json`; derived metrics mis_z=7.38, pLI≈0, LOEUF=0.66. Outcome: **succeeded**.

2. **ClinVar record counts** — Input: `MYH7[gene]` + disease free-text/controlled terms → Method: eutils esearch → Output: `data/clinvar_MYH7_counts.json`; 6011 total, 1479 P/LP, ~84% cardiomyopathy. Outcome: **succeeded** (with a documented vocabulary artifact).
3. **GenCC / ClinGen gene-disease validity** — Input: MYH7 → Method: web/database lookup → Output: none. Outcome: **failed / could not be verified**. Reported explicitly as a source-level absence; **no silent fallback** — the SRX conclusions do not depend on this source.

No patient-level, controlled-access, or large raw-omics datasets were downloaded. No wet-lab or structural (Phenix) analysis was run, as this hypothesis search is literature- and public-database-driven. The computational outputs above are small derived tables/JSON suitable for the artifact bundle under `kb/hypotheses/MYH7-Related_Skeletal_Myopathy/rod_srx_energetics/openscientist_artifacts`. The bundle checksums were verified in iteration 5 (inputs/outputs match MANIFEST).

Limitations and Knowledge Gaps

1. **Missing causal step: ATP demand → weakness in patients.** The decisive link is inferred, not measured, in MYH7 patients. *What was checked:* PubMed for MYH7-specific in-vivo energetics (e.g., ^{31}P-MRS) in genotyped patients — none found linking SRX-driven ATP demand to clinical fatigue. *Resolution:* ^{31}P-MRS / near-IR energetics in genotyped Laing patients vs controls. (General ^{31}P-MRS fatigue/energetics studies exist — e.g. PMID: 21719720, PMID: 12914557 — but none in MYH7 cohorts.)
2. **Hypercontractility discordance (mouse vs human).** Conflicting across systems: mouse hypercontractile, human contractility unchanged (PMID: 37788100 vs PMID: 38690726). Matters because the seed label implies a contractile component. *Resolution:* standardized myofiber mechanics across more human variants and the mouse under matched conditions.
3. **Causal primacy vs aggregation.** For R1500P, SRX loss and dominant-negative aggregation co-occur (PMID: 22155079). Which is primary — or whether aggregation is upstream of SRX loss — is unknown. *Resolution:* temporal/perturbation studies (e.g., does SRX-stabilizing MYK-581 rescue weakness without clearing aggregates?).
4. **Non-specificity of SRX loss.** SRX disruption + raised resting ATP also occurs in NEB nemaline myopathy (PMID: 36528760) and inversely in T2DM (PMID: 40295335). The node is convergent, not MYH7-specific.

5. **Assay-equivalence caveat.** Structural (X-ray order) and biochemical (mantATP) readouts are not interchangeable (PMID: 36696446, PMID: 38103642); the seed papers treat them as one.
6. **Source/data absences.** GenCC and ClinGen gene–disease validity could not be verified this run (labeled unverified, not "absent"). ClinVar shows the skeletal phenotype is heavily under-represented (~16% of records), limiting cohort-scale validation.
7. **Small human n.** The pivotal human biopsy study rests on a limited number of patients/variants; generalization across the MYH7 rod-variant spectrum is unestablished.

Alternative Models

Model	Relation to seed hypothesis	Support
Dominant-negative aggregation / WT sequestration	**Alternative or upstream** — aggregates could disrupt filaments before/independent of SRX loss	PMID: 22155079, PMID: 22918376
Thick-filament assembly / coiled-coil defect	**Upstream cause** — perturbed backbone could produce BOTH SRX loss and aggregation	PMID: 22918376, PMID: 37788100
Convergent SRX-hub dysregulation	**Parallel/non-specific** — SRX loss as a shared endpoint of many sarcomeric lesions	PMID: 36528760, PMID: 19966283
Hypercontractility-driven damage (mouse)	**Downstream consequence**, possibly species-specific	PMID: 38690726
Energetic stress → fatigue (energetic-only, no contractile change)	**Refinement of the seed** matching the human data	PMID: 37788100

Discriminating Tests

1. **In-vivo patient energetics → weakness (highest priority).** ^{31}P-MRS (and near-IR) in genotyped MYH7 rod-variant patients vs matched controls, correlated with quantitative strength/fatigability. *Expected if hypothesis true:* elevated resting Pi/ATP turnover tracking with weakness. Distinguishes the energetic model from purely structural models.
2. **SRX-stabilizer rescue decoupled from aggregation.** Treat R1500P models/patient myofibrils with MYK-581 (or mavacamten-class) and test whether strength/fatigue normalizes *without* clearing aggregates. *Expected if energetics is primary:* functional rescue despite persistent inclusions. Directly separates F001 from F007.
3. **Cross-variant myofiber mechanics panel.** Standardized mantATP + X-ray + mechanics across diverse rod vs head-domain MYH7 variants. *Expected:* rod variants cluster on SRX loss; head variants may differ — maps the hypothesis's subtype scope.
4. **Temporal ordering in iPSC-myotubes.** Live imaging of an SRX reporter vs aggregate formation across differentiation. *Expected if aggregation is upstream:* inclusions precede SRX loss.
5. **Orthogonal SRX quantitation.** Pair mantATP with an independent SRX readout to address the assay-equivalence caveat (PMID: 38103642).

Curation Leads (require curator verification)

Candidate evidence references & snippets: - PMID: 37788100 — *"basal myosin ATP consumption was increased and the myosin super-relaxed state was decreased in vivo"* (SUPPORT, human) **and** *"myofiber contractility was not affected"* (QUALIFY — add as explicit qualifier node). - PMID: 38690726 — *"a conformational shift of the myosin heads toward the more reactive disordered relaxed (DRX) state, causing hypercontractility and greater ATP consumption"* (SUPPORT, mouse; tag hypercontractility as species-specific). - PMID: 22155079 — *"aggregates ... that sequester WT myosin"* (COMPETING — add dominant-negative edge for R1500P). - PMID: 36528760 — *"super-relaxed state, was significantly impaired, inducing an increase in the energy (ATP) consumption of resting muscle fibres"* (QUALIFY — convergence node, NEB). - PMID: 22918376 — *"altered biochemical and biophysical properties of the light meromyosin, which is essential for thick filament assembly"* (COMPETING, review-level).

Candidate pathophysiology nodes/edges: add node "SRX loss (convergent hub)"; add qualifier edge "human: energetic without contractile change"; add competing edge "rod variant → aggregation → WT sequestration"; mark edge "ATP demand → weakness (human)" as INFERRED/unconfirmed.

Candidate ontology terms: skeletal muscle myofiber (CL:0000188); myosin II complex / thick filament (GO:0032982); ATP metabolic process (GO:0046034); muscle filament sliding / super-relaxed state (no precise GO term — flag ontology gap).

Candidate status/subtype restrictions: keep status **EMERGING**; restrict confident claims to the **energetic arm**; flag hypercontractility as **mouse-restricted**; add subtype note that evidence is strongest for Laing distal myopathy (rod/LMM), sparse elsewhere.

Candidate knowledge_gaps / discussion prompts: (a) no MYH7-specific in-vivo patient energetics-to-weakness study found (search date 2026-10-16); (b) causal primacy SRX-loss vs aggregation unresolved for R1500P; (c) GenCC/ClinGen validity **unverified** this run (not confirmed absent); (d) ClinVar skeletal under-representation (~16%).

Evidence Base Summary

- **Supporting the energetic arm (2 primary studies):** PMID: 38690726 (mouse), PMID: 37788100 (human) — plus foundational PMID: 19966283 and computational premise support (gnomAD).
- **Qualifying scope/specificity (4):** PMID: 37788100 (no contractility change), PMID: 36528760 (NEB convergence), PMID: 40295335 (T2DM opposite direction), PMID: 36696446 / PMID: 38103642 (assay caveats).
- **Competing mechanisms (2):** PMID: 22155079 (dominant-negative), PMID: 22918376 (assembly/aggregation).
- **Clinical context (1):** PMID: 24664454 (phenotype + explicit mechanistic gap).

Overall: 2 of 2 systems that directly measured it confirm SRX loss + raised basal ATP (the hypothesis's core). 1 of 1 human study that measured contractility found no change (qualifying the hypercontractility claim). At least 1 structurally unrelated gene reproduces the SRX phenotype (non-specific). At least 1 established competing mechanism exists for the same allele. The energetic mechanism is real and well-measured; its causal primacy and human relevance to weakness are unresolved.

Report generated after 5 investigation iterations; 8 findings confirmed; 17 papers reviewed; 2 computational database queries succeeded; 2 database lookups (GenCC, ClinGen) could not be verified. Artifact bundle checksums verified (iteration 5). Search date: 2026-10-16.